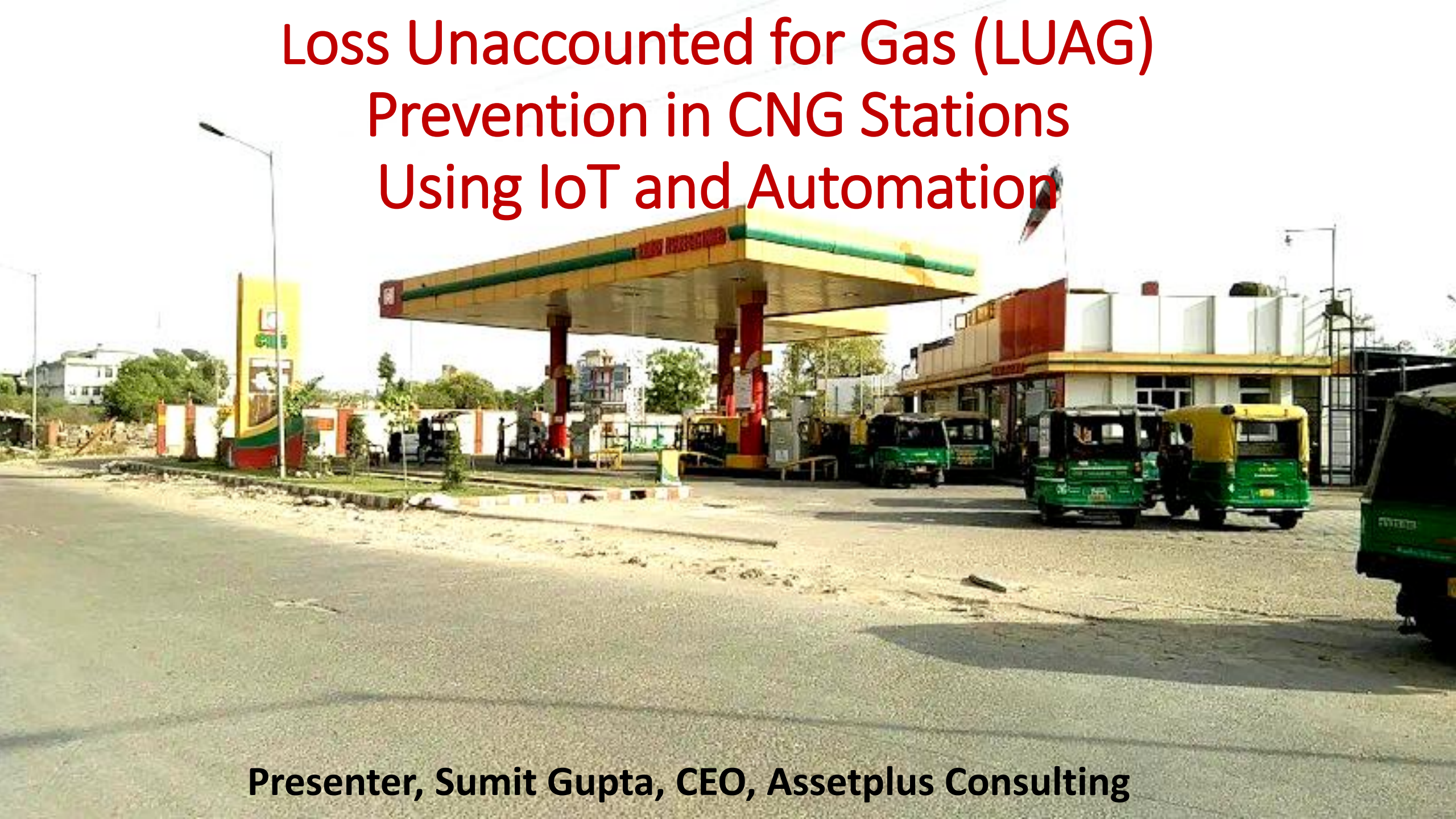
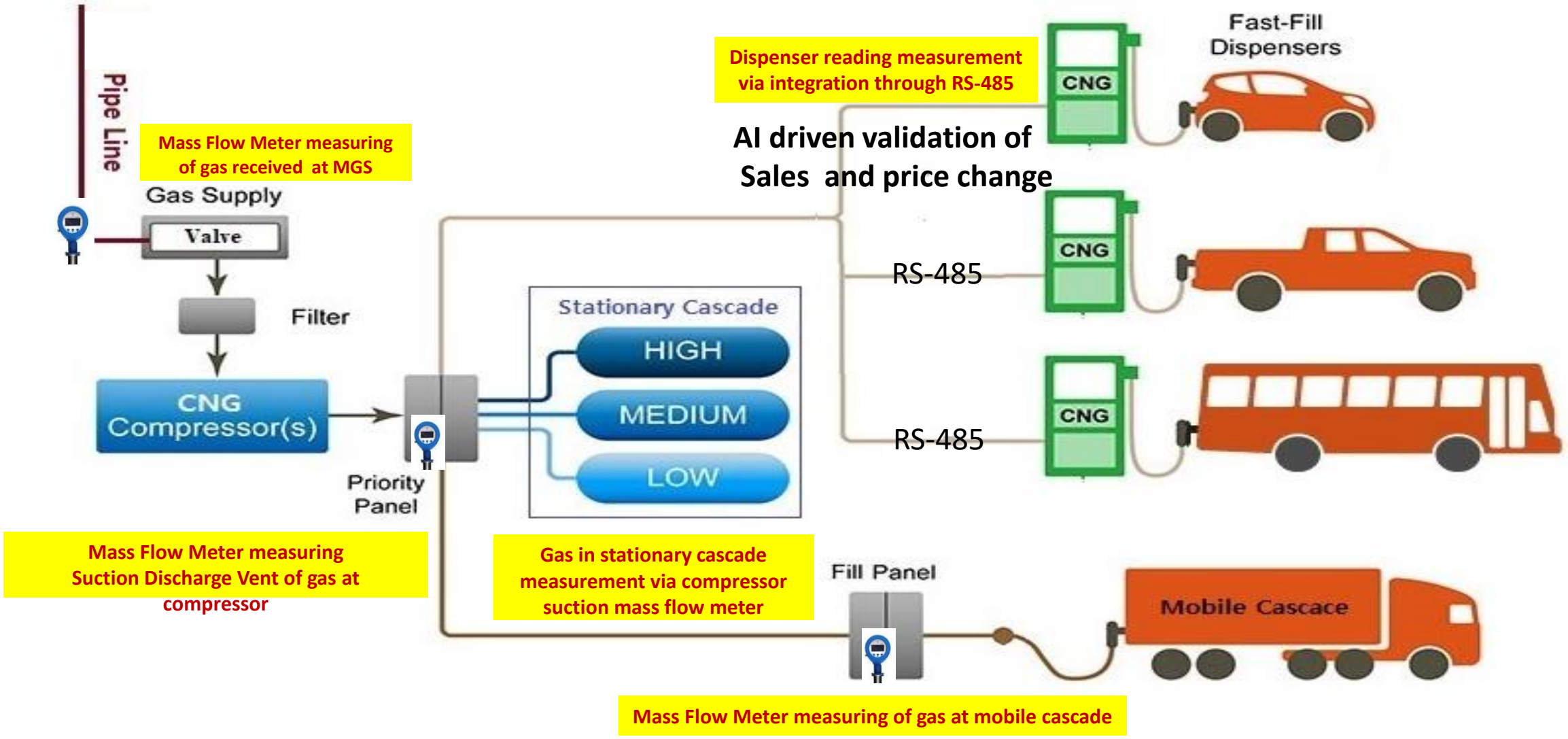
Loss Unaccounted for Gas (LUAG) Prevention in CNG Stations Using IoT and Automation



Presenter, Sumit Gupta, CEO, Assetplus Consulting

INTRODUCTION - CNG MOTHER GAS STATION



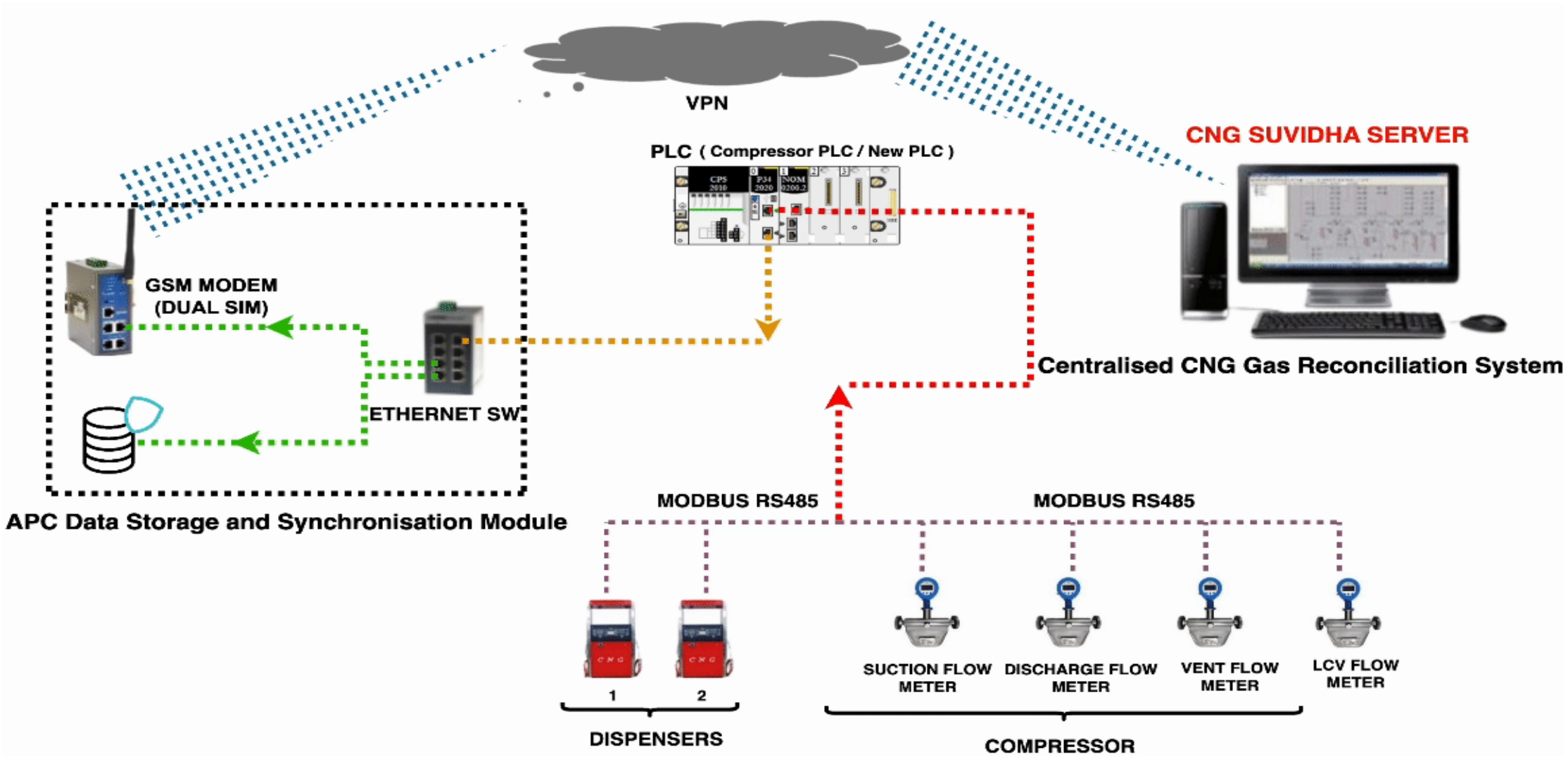
IOT –AI system over traditional SCADA

- City Gas Distribution entities have been traditionally making use of expensive , bulky , time consuming **SCADA system** for an integrated data driven control of operation and **Network Gas Reconciliation** leading to measurement of **GAS LOSS** (LUAG – Loss unaccounted for Gas).
- Developing City Gas Distribution network is **CAPEX intensive** project , where in the new geographical areas , instead of investing into expensive SCADA system for GAS reconciliation/ gas Los reduction , the **innovative IOT-AI based quick to implement system** was tried and implemented at **RAJASTHAN STATE GAS LIMITED (RSGL),** a JV of GAIL GAS LIMITED and RAJASTHAN STATE Govt, with successful outcomes

- **Dispenser Data Capture** – Automated collection of dispensing data in real time and remote price change.
- **Compressor Data Integration** – Monitoring and recording compressor parameters for operational tracking.
- **LCV Filling Point Data** – Capturing and reconciling LCV filling transactions.
- **LCV Sales Challan Generation** – Automatic generation of sales challans for LCVs based on authenticated data.
- **POS Payment Mechanism** – Integration with POS systems for seamless billing and payment tracking.
- **Gas Reconciliation** – Comprehensive reconciliation of dispenser, compressor, and LCV data to determine actual consumption and station loss.

IOT system implementation

IOT station system communication &Control Architecture



IMPACT of IOT based CNG Station automation over SCADA

- **Reduction In Manpower**

  Earlier, 3 personnel (1 per shift) were deployed at each CNG station for:

  - Hourly meter readings
  - LCV challan generation
  - Manual register maintenance and record keeping
- With automation and online system implementation, these activities are now fully digitized, eliminating the need for dedicated manpower for these tasks.

💰 **Cost Savings (Per Station)**

  - Monthly Saving: ₹20,000 per person × 3 persons = ₹60,000 per month
  - Annual Saving: ₹60,000 × 12 = ₹7.2 lakh per year
- **Impact:**
  - Reduced operational cost • Improved efficiency • Better utilization of workforce for core operations

CASE STUDY -RAJASTHAN STATE GAS LIMITED, KOTA

- **Reduction in Gas Loss at CNG Stations**
  - Gas loss at CNG stations reduced from 1.29% to 0.23%, achieving a net reduction of 1.06%.
  - With average daily CNG sale of 12,700 kg, gas saving is : 12,700 × 1.06% = 134.62 kg per day
  - Daily Financial Saving: 134.62 kg × ₹92.5 per kg = ₹12,452 per day
  - Monthly Saving: ₹12,452 × 30.5 days ≈ ₹3.8 lakh per month
  - Annual Saving (Per Station): ≈ ₹45.5 lakh per year
- **Projected Savings Across 19 CNG Stations**
  - Currently operating 19 CNG stations within our GA.
  - Monthly Saving: 19 × ₹3.8 lakh = ₹72.2 lakh per month
  - Annual Saving: ₹72.2 lakh × 12 ≈ ₹866.4 lakh per year (₹8.66 crore annually)

KEY TAKEAWAYS / RECOMMENDATIONS

📊 Real-Time Monitoring & Performance Optimization

- Live tracking of compressor running hours, fuel intake, exhaust temperature, and efficiency
- Instant monitoring of breakdowns, downtime, and gas losses
- Data-driven analysis to improve operational performance

🛠 Smart Maintenance & Digital Records

- Complete online maintenance history of dispensers and compressors
- Real-time tracking of breakdowns and service activities
- Enables preventive maintenance planning, Improves transparency and operational control

🔄 Intelligent Automation & Control

- Automatic dispensing data recording with real-time monitoring
- Remote online price change reduces manpower dependency
- POS integration ensures seamless billing and payment tracking

🔐 Revenue Protection & Data Integrity

- 24×7 monitoring of raw site data, Eliminates manual data manipulation
- Strengthens theft detection and revenue assurance

RSGL Control Room , Kota (Rajasthan)

THANKS